# T-05 - Retrieval-Augmented Generation and the Study of Hallucination

**Phase 1 Report - Topic, Dataset, Baseline, and Evaluation Plan**

**Narrow project question**
Can a lightweight RAG pipeline using dense retrieval reduce unsupported or hallucinated answers compared with the same generator used without retrieval on a small SQuAD 2.0 subset?

## 1. Project Overview and Scope

This project prepares a small, reproducible RAG baseline for studying hallucination in an educational question-answering setting. The comparison is intentionally controlled: RAG and No-RAG use the same generator, and the only experimental difference is whether a retrieved passage is supplied. Phase 1 defines the task, documents the data, implements the baseline, and specifies the evaluation and error-analysis plan. Final metric results belong to Phase 2.

| Phase 1 element | Decision |
|---|---|
| Topic | T-05 - Retrieval-Augmented Generation and Hallucination |
| Research design | Controlled comparison: RAG vs. No-RAG |
| Training | None; pretrained off-the-shelf models only |
| Runtime target | Free Google Colab, CPU-compatible |
| Out of scope | Fine-tuning, training from scratch, custom loss, UI, deployment, formal statistics |

## 2. Dataset

| Item | Description |
|---|---|
| Dataset | SQuAD 2.0 (Hugging Face: rajpurkar/squad_v2) |
| Split | train, after deterministic shuffling with SEED = 42 |
| Working subset | Approximately 300 unique contexts and 80 questions |
| Question mix | Answerable and unanswerable; target is about 25% unanswerable |
| Core fields | context, question, answers["text"], answers["answer_start"] |
| Unanswerable rule | len(example["answers"]["text"]) == 0 |

Why this dataset fits T-05: SQuAD 2.0 contains both answerable and deliberately unanswerable questions. It therefore supports analysis of two central RAG failures: missing evidence and generation that ignores or exceeds available evidence.

# 3. Dataset Preparation and Baseline Architecture

## 3.1 Deterministic subset construction

- Shuffle the SQuAD 2.0 train split with SEED = 42.
- Collect the first 300 unique contexts after shuffling.
- Gather questions attached to those contexts.
- Select about 80 questions while reserving about 20 unanswerable examples.
- Keep the same subset and index for both RAG and No-RAG evaluation.

**Design rationale**
The corpus is deliberately small so retrieval, generation, and qualitative inspection remain understandable and reproducible on a free Colab runtime. The subset is suitable for a course baseline, but not for population-level claims.

## 3.2 Baseline components

| Role | Tool / model | Purpose |
|---|---|---|
| Dataset loader | Hugging Face datasets | Load SQuAD 2.0 and select the subset |
| Embedding model | sentence-transformers/all-MiniLM-L6-v2 | Encode questions and contexts into 384-d vectors |
| Vector search | FAISS IndexFlatL2 | Exact nearest-neighbour ranking over 300 context vectors |
| Generator | google/flan-t5-small | Generate a short answer from the prompt |
| No-RAG baseline | Same FLAN-T5-small | Answer the question without retrieved evidence |

**Question** -> **MiniLM embedding** -> **FAISS top-1** -> **FLAN-T5 answer**

In RAG mode, the generator receives the top-ranked retrieved context. In No-RAG mode, the same generator receives only the question. This isolates the effect of retrieval and avoids comparing different language models.

## 3.3 Reproducibility choices

- Fixed random seed for Python and NumPy.
- Pinned public model identifiers and a fixed FAISS index type.
- L2-normalised embeddings and exact flat search.
- Greedy decoding (do_sample=False, num_beams=1).
- Single notebook that installs dependencies and runs from top to bottom.

## 4. Baseline Implementation

The notebook follows a low-code, linear workflow. Each stage is kept visible so the team can explain the entire pipeline during the oral review.

| Step | Notebook action |
|---|---|
| 1 | Install datasets, sentence-transformers, faiss-cpu, and transformers. |
| 2 | Load SQuAD 2.0 train split and inspect schema. |
| 3 | Build the deterministic, mixed answerable/unanswerable subset. |
| 4 | Embed all selected contexts with all-MiniLM-L6-v2. |
| 5 | Create FAISS IndexFlatL2 and add the context vectors. |
| 6 | Define retrieve(question, k) for ranked context retrieval. |
| 7 | Load FLAN-T5-small and define a deterministic generation helper. |
| 8 | Define RAG and No-RAG generation functions. |
| 9 | Run a three-question smoke test; do not treat it as evaluation. |

### 4.1 Prompt design

**RAG prompt**
Answer the question using the context. If the context does not contain the answer, say: I don't know.

Context: <retrieved passage>
Question: <question>
Answer:

**No-RAG prompt**
Answer the question.

Question: <question>
Answer:

The RAG prompt explicitly permits abstention. The No-RAG prompt is intentionally minimal because it is a lower-bound comparison, not a second engineered system. Both modes use the same maximum generation length and deterministic decoding.

### 4.2 Phase 1 sanity check

Three examples are used only to verify that loading, retrieval, prompting, and generation run end-to-end: two answerable questions and one unanswerable question. No aggregate metrics or final conclusions are reported in Phase 1.

**Correct interpretation**
A successful smoke test proves that the baseline is wired correctly. It does not establish that RAG improves accuracy or reduces hallucination; those claims require the planned Phase 2 evaluation.

## 5. Evaluation Plan for Phase 2

The project specification asks for a small number of task-appropriate metrics. The plan therefore distinguishes primary metrics from supporting diagnostics.

| Component | Metric | Interpretation |
|---|---|---|
| Retrieval - primary | Recall@5 | Fraction of answerable questions whose gold context appears in the top five results. |
| Retrieval - support | MRR | Average reciprocal rank of the gold context; rewards higher placement. |
| Generation - primary | Token F1 | Token overlap with the best gold answer; gives partial credit. |
| Generation - support | Exact Match | Strict normalised answer equality. |
| Hallucination | Unanswerable abstention rate | Fraction of unanswerable questions for which the model declines to answer. |

RAG and No-RAG will be evaluated on the same questions. For questions with multiple gold answers, the maximum score across all accepted references should be used. Unanswerable questions are identified from an empty answers["text"] list.

## 6. Simple Analysis Plan

| Category / slice | Question answered by the analysis |
|---|---|
| Retrieval miss / absent evidence | Did the generator fail because the gold passage was not retrieved? |
| Ignored evidence | Was the correct passage retrieved but the answer still unsupported or wrong? |
| Unanswerable failure | Did the model invent an answer instead of abstaining? |
| Short vs. long questions | Does question length change retrieval or answer quality? |

Representative examples will be shown in a compact table containing the question, gold answer, retrieved passage, RAG answer, No-RAG answer, error type, and a one-sentence explanation. The goal is interpretability, not formal statistical testing.

### Planned decision logic

- Gold context absent from the retrieved set + wrong content answer -> retrieval-miss / absent-evidence hallucination.
- Gold context retrieved + wrong content answer -> ignored-evidence hallucination.
- SQuAD marks the question unanswerable + model gives content answer -> unanswerable failure.
- Model explicitly declines to answer -> abstention.

# 7. Scope, Limitations, and Deliverables

## 7.1 Known limitations

- Small corpus and question sample: later results will be indicative rather than statistically conclusive.
- FLAN-T5-small is intentionally lightweight and may have weak factual recall and poor abstention behaviour.
- The prompt is simple; no reranker, hybrid retrieval, prompt search, fine-tuning, or confidence calibration is used.
- Flat L2 search is appropriate for 300 vectors but does not test large-scale retrieval efficiency.
- Automatic lexical metrics may underrate paraphrases; hallucination labels will still require limited human inspection.
- The selected corpus contains each evaluated question's source context by construction, so retrieval evaluation is closed-corpus and optimistic relative to a real deployment.

## 7.2 Phase 1 deliverables checklist

| Required item | Status | Evidence |
| --- | --- | --- |
| Narrow project question | Done | Section 1 and notebook overview |
| Dataset description | Done | Source, subset, fields, and answerability rule documented |
| Baseline setup | Done | Executable Colab notebook with fixed models and index |
| Evaluation plan | Done | Retrieval, generation, and abstention metrics defined |
| Simple analysis plan | Done | Error categories and data slice listed |
| Phase 1 report <= 6 pages | Done | This five-page report |

## 7.3 External AI tools acknowledgement

An external AI assistant was used for drafting, debugging, and report organisation. The team remains responsible for understanding and defending every design choice, code cell, and planned metric. No synthetic data, fabricated results, or invented citations are included.

# 8. Conclusion

Phase 1 establishes a complete and appropriately lightweight baseline for T-05. The dataset is relevant, the controlled RAG/No-RAG comparison is reproducible, and the evaluation plan separates retrieval failure from generation failure. The notebook is therefore ready for Phase 2, where the same baseline will be evaluated without changing the dataset or model configuration.

**Submission note**
Companion notebook: T05_Phase1_LowCode_RAG.ipynb. Run all cells from the beginning in Google Colab to reproduce the Phase 1 baseline.